

Travaux pratiques (TP)

Partie I $\rightarrow$ Théorie

$$1-a) \frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial b} ; \mathcal{L}(w, b) = \frac{1}{2N} \sum_{i=1}^N (w x_i + b - y_i)^2$$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{1}{N} \sum_{i=1}^N x_i (w x_i + b - y_i)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{1}{N} \sum_{i=1}^N (w x_i + b - y_i)$$

$$b) \nabla \mathcal{L}(w^*, b^*) = 0$$

$$\nabla \mathcal{L}(w^*, b^*) = \begin{pmatrix} \frac{1}{N} \sum_{i=1}^N (w^* x_i + b^* - y_i) x_i \\ \frac{1}{N} \sum_{i=1}^N (w^* x_i + b^* - y_i) \end{pmatrix}$$

$$\nabla \mathcal{L}(w^*, b^*) = 0 \Rightarrow \begin{cases} \sum_{i=1}^N (w^* x_i + b^* - y_i) x_i = 0 \text{ (1)} \\ \sum_{i=1}^N (w^* x_i + b^* - y_i) = 0 \text{ (2)} \end{cases}$$

$$\frac{1}{N} \sum_{i=1}^N x_i = \bar{x} ; \quad \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}$$

$$(2) \Rightarrow \sum_{i=1}^N (w^* x_i + b^* - y_i) = 0$$

$$\Rightarrow w^* \sum_{i=1}^N x_i + \sum_{i=1}^N b^* - \sum_{i=1}^N y_i = 0$$

$$\Rightarrow \frac{1}{N} w^* \sum x_i + \frac{1}{N} \sum b^* - \frac{1}{N} \sum y_i = 0$$

$$\Rightarrow w^* \bar{x} + b^* - \bar{y} = 0$$

$$\Rightarrow b^* = \bar{y} - w^* \bar{x} \quad (3)$$

$$\sum (w^* x_i + b^* - y_i) x_i = 0 \Rightarrow \sum w^* x_i^2 + b^* \sum x_i - \sum x_i y_i = 0$$

$$\Rightarrow w^* \sum x_i^2 + b^* \sum x_i - \sum x_i y_i = 0$$

$$\Rightarrow w^* \sum x_i^2 + b^* N \bar{x} - \sum x_i y_i = 0$$

$$\Rightarrow w^* \sum x_i^2 + (\bar{y} - w^* \bar{x}) N \bar{x} - \sum x_i y_i = 0$$

$$\Rightarrow w^* (\sum x_i^2 - N \bar{x}^2) = \sum x_i y_i - N \bar{x} \bar{y}$$

$$\Rightarrow N w^* \left(\frac{1}{N} \sum x_i^2 - \bar{x}^2 \right) = N \left(\frac{1}{N} \sum x_i y_i - \bar{x} \bar{y} \right)$$

$$w^* \underbrace{\left(\frac{1}{N} \sum x_i^2 - \bar{x}^2 \right)}_{\text{Var}(x)} = \underbrace{\frac{1}{N} \sum x_i y_i - \bar{x} \bar{y}}_{\text{Cov}(x, y)}$$

$$w^* \text{Var}(x) = \text{Cov}(x, y) \Rightarrow \boxed{w^* = \frac{\text{Cov}(x, y)}{\text{Var}(x)}}$$

[MCD]

$$2) f(x, y) = x^2 e^y - 3xy^2$$

$$a) \nabla f(x, y) = \begin{pmatrix} 2xe^y - 3y^2 \\ x^2 e^y - 6xy \end{pmatrix}$$

$$b) \nabla^2 f(x, y) = \begin{pmatrix} 2e^y & 2xe^y - 6y \\ 2xe^y - 6y & x^2 e^y - 6x \end{pmatrix}$$

$$c) A = (1, 0); \quad \nabla f(A) = ?; \quad \nabla^2 f(A) = ?$$

$$\nabla f(1, 0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}; \quad \nabla^2 f(1, 0) = \begin{pmatrix} 2 & 2 \\ 2 & -5 \end{pmatrix}$$

$$3) g(x) = \|x - a\|^2 \quad x^T x = \|x\|^2$$

$$a) \text{ Montrons que } \nabla g(x) = 2(x - a)$$

$$g(x) = (x - a)^T (x - a) = x^T x - 2a^T x + a^T a$$

$$f(x) = x^T x + b^T x + c \quad \left| \begin{array}{l} b = -2a \\ c = \|a\|^2 \end{array} \right.$$

$$\nabla f(x) = 2I x + b$$

$$\nabla g(x) = 2I x - 2a$$

$$\underline{\underline{\nabla g(x) = 2(x - a)}}$$

$$g(x) = (x - a)^2$$

$$\nabla g(x) = 2x(x - a)$$

$$\nabla g(x) = 2(x - a)$$

b) Point critique.

$$\nabla g(x) = 0 \Leftrightarrow 2(x-a) = 0$$

$$\Leftrightarrow x - a = 0$$

$$\Leftrightarrow x = a, \quad \left. \begin{array}{l} x \in \mathbb{R}^n \\ a \in \mathbb{R}^n \end{array} \right\}$$

$$\underline{x^* = a}$$

c) $\nabla^2 g(x)$

$$\nabla g(x) = 2(x-a) \Rightarrow \nabla^2 g(x) = 2 \cdot I_n$$

Les valeurs propres de $\nabla^2 g(x)$ sont > 0 et donc la Hessienne est définie positive $\rightarrow$ fonction convexe

• Point critique d'une fonction convexe = minimum global

4) $h(x) = x^4$; $h'(x) = 4x^3$; $h''(x) = 12x^2$

$h''(x) \geq 0$ / strictement convexe sauf en un point isolé.

$$\mathbb{C}^2 \rightarrow \mathbb{C}^2$$

$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$